PHASE-A RESULTS: THE FREE EMA-KL SURROGATE PREVENTS STALENESS COLLAPSE, AND ITS THEORY PICKS THE RIGHT DOSE

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Abstract. Asynchronous RL for LLMs trains on rollouts from stale policy snapshots, and modern recipes increasingly drop KL regularization entirely. A companion theory note shows that a KL penalty anchored on the sampler's cached log-probabilities, $c\,(\log T_n - \log T_{n-\Delta})$ with $c = \alpha/(\Delta(1-\alpha))$, is a zero-cost first-order surrogate for KL regularization against an implicit EMA of the trainer. We test it on Qwen3-4B/Countdown (TBA, no centering) under high and heterogeneous staleness, with and without NVFP4-quantized rollouts. (1) Staleness alone breaks unregularized training — homogeneous Δ=32 falls from eval 0.75 to 0.14 by step 200, and a heterogeneous mixture $\Delta_i \in \{1, 4, 10, 32\}$ collapses ∼60 steps later (0.75 → 0.002 by step 300) — while every arm with the surrogate at the theory-prescribed strength is stable, reaching eval 0.79–0.83. (2) Per-rollout calibration $c_i = \alpha/(\Delta_i(1-\alpha))$ ties the best uniform dose found by a 3× sweep (peaks 0.816 vs. 0.817); that best dose is itself the one the theory transports from a known-good run via the βc invariance, and tripling it costs ∼1.5 points — the theory selects the right strength without a sweep. (3) The surrogate's relative error is strongly Δ-dependent as predicted (5.9 at $\Delta \leq 2$ vs. 0.19 at $\Delta \geq 21$), yet training is unaffected. (4) NVFP4 quantization compounds with staleness: the FP4 Δ=32, β=0 arm collapses ∼70 steps earlier than its BF16 twin from a lower peak, while $\Delta \lesssim 10$ stays stable; the same theory dose $\beta c = 0.0045$ keeps Δ=10 healthy and extends the Δ=32 collapse boundary ∼3×, though not to full-horizon stability. In the FP4-heterogeneous cell that motivated this study, the β=0 control collapses at ∼step 205 while both KL arms train through their full horizons; per-rollout and uniform dosing tie on eval (peaks 0.771 vs. 0.764, single seed) — the rescue claim holds, and the weighting refinement is not separable from a well-chosen constant. (5) Two further experiments are reported *in progress*, at roughly a third of their intended horizon and read at a common step floor. Mixing the same lags *within* each K-group — the only regime in which per-rollout weighting can act — moves the β=0 collapse ∼40–55 steps earlier in both precisions and lifts the calibrated-vs-uniform comparison out of the tie band (±0.004) in both, but with opposite signs (+0.029 eval for per-rollout in BF16, −0.010 under NVFP4); the structural prediction — allocation is inert only while groups are age-homogeneous — survives, the ranking does not yet resolve. (6) An error-triggered switch from the surrogate to the exact EMA-KL term, designed to catch the FP4 Δ=32 late failure, fires where intended (step 262, before any eval turn, from a trajectory indistinguishable from its twin) but brings the failure ∼60–80 steps forward: a free accuracy monitor is a usable alarm, and swapping estimators mid-run is not a free correction.

1. Introduction

Asynchronous RL pipelines for LLMs decouple generation from optimization: inference workers produce rollouts from a policy snapshot that is Δ steps behind the trainer, and the trainer consumes them after the fact. As throughput pressure pushes Δ up — and as quantized inference adds a second source of trainer/sampler mismatch — the question of whether and how to regularize against this drift becomes central. Current practice pulls in two directions: explicit KL regularization against a reference policy costs an extra reference forward pass per batch, so several prominent recipes simply remove the KL term [8], including the humans& NVFP4 reinforcement-learning recipe, which manages mismatch through systems-level bit-exact contracts and trains KL-free [7].

A companion theory note [1] observes that a middle path is already free. In an asynchronous pipeline the sampler's log-probabilities $\log T_{n-\Delta}$ are cached for importance sampling, so the penalty $c\,(\log T_n - \log T_{n-\Delta})$ costs nothing extra — and, under a locally-linear-drift assumption,

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it is a first-order surrogate for KL regularization against an *exponential moving average* of the trainer policy, provided the coefficient is calibrated as $c = \alpha/(\Delta(1-\alpha))$. The calibration makes two testable predictions: rollouts with different lags Δ_i should receive different weights c_i to maintain a constant effective regularization strength, and a known-good effective dose βc can be transported across staleness regimes without re-tuning.

This report evaluates the surrogate empirically through seven questions, each answered by one figure: (i) does staleness alone break unregularized asynchronous RL, and does the free surrogate prevent it (Fig. 1: yes, and yes); (ii) does per-rollout calibration beat a well-chosen constant dose (Fig. 2: a tie when K-groups are age-homogeneous, but the constant itself is the one the theory prescribes); (iii) is the first-order surrogate accurate everywhere (Fig. 3: no — its error is strongly Δ-dependent, as the theory warns — yet training is robust to it); (iv) does quantization change the picture (Fig. 4: yes — NVFP4 rollouts compound with staleness, moving the Δ=32 collapse boundary from ∼step 170 to ∼100; the same theory dose fully rescues Δ=10 and delays, but does not eliminate, the Δ=32 failure); (v) does staleness weighting matter under FP4 heterogeneity (Fig. 5: the rescue holds; weighting vs. a constant is a tie at one seed); (vi) does mixing policy ages *within* each K-group — the regime the tie in (ii) and (v) was blamed on — change that answer (Fig. 6: it ends the tie, and makes the unregularized failure worse, but the two precisions disagree on the sign at this horizon); (vii) can a free, error-triggered switch to the exact EMA-KL term rescue the one arm the surrogate fails to hold (Fig. 7: no — the alarm fires correctly and the intervention hurts). Questions (vi) and (vii) are answered by runs still in flight and are marked in progress throughout. Except where noted, results are single-seed; tie bands and caveats are stated with each finding.

2. Setup

All runs use Qwen3-4B-Instruct-2507 on Countdown with TBA (no centering, importance sampling on), lr 10^{-6}, K=16, seq. length 4096, 720 target steps, eval every 20 steps, $\alpha = 0.9$, on 2 nodes (4×H100 inference / 4×H100 training) with 24 h wall-clock allocations (curves end between steps 490 and 690). Full configurations are in Appendix A.

Heterogeneous staleness. `--rl.staleness-offsets` $o_{0..3}$ pins each of 4 data-parallel vLLM ranks to a fixed checkpoint lag (reload every step to step $- o_r$). K-groups stay policy-homogeneous; training batches pool ranks and therefore mix policy ages (verified in the rollout parquets: the trainer's step-10 batch carries generating checkpoints $\{9, 6, 0, 0\}$). The homogeneous arms use offsets 10, 10, 10, 10 (reference) or the legacy reload rule at `async_level`=32 (β=0 control).

Within-group age mixing. `--rl.staleness-within-group` replaces the per-rank pinning with rounds-based generation: each prompt's K samples are produced in one round per offset (oldest checkpoint first) and the rounds are merged before reward and advantage centering, so every K-group spans all four ages $\Delta_i \in \{1, 4, 10, 32\}$. Parquet rows carry the per-output generating checkpoint step; the trainer's per-token Δ_i machinery is unchanged. The offset multiset, `async_level`, and doses are identical to the between-group arms — only the placement of the ages differs, which is what isolates the allocation question.

Runs in progress, and the common-step-floor rule. Sections 3.6 and 3.7 report seven runs that have not reached their 720-step target; their last completed evals span steps 240–360, and arms within a comparison sit at different steps. Reading each arm at its own latest step would confound arm with horizon, so every cross-arm statement in those sections is made at the *common step floor* — the minimum last-eval step over the arms compared (step 240 for the 2×3 grid, step 360 for the fallback pair) — and paired statistics are computed only over steps both arms have reached. Peaks from these runs are quoted with their step and are not comparable to the completed runs above.

Anchors and calibration. Every arm anchors both the importance ratio and the KL term on the sampler's own vLLM logprobs. Per-rollout calibration uses each sample's true lag: $c_i = \alpha/(\Delta_i(1-\alpha))$, $\Delta_i \geq 1$; the uniform arms apply one dose $w = \beta\, c_{\text{glob}}$ to every rollout ($c_{\text{glob}} = 0.9/3.2 = 0.28125$). The per-token clamp is in raw log-ratio units (±10 before the coefficient), so truncation is Δ-invariant.

FP4 emulation. FP4 arms use symmetric NVFP4 QDQ emulation: STE on trainer forwards, QDQ on sampler checkpoints, bit-exact between the two (252 patched linears; embeddings, `lm_head`, and norms stay BF16). The one exception is the asymmetric arm (BF16 trainer, QDQ sampler), reported as a negative result in Section 3.4.

How the uniform constant was chosen (design decision). The uniform arms' primary constant was fixed by *transporting* the known-good effective strength from the homogeneous Δ=10 baseline ($\beta c = 0.005 \times 0.9 = 0.0045$) and anchoring the global coefficient at the *maximum* staleness: c_{glob} uses $\Delta_{\max} = 32$, and $\beta = 0.0045/0.28125 = 0.016$ so that $\beta\, c_{\text{glob}} = 0.0045$. This choice does *not* account for the fact that most lags in the $\{1, 4, 10, 32\}$ mixture are below 32: because the raw log-ratio magnitude grows roughly linearly in Δ_i, a constant raw weight yields an effective per-rollout dose $\propto \Delta_i/32$ — sub-maximal lags are effectively under-regularized relative to the calibrated arm, whose effective dose is flat at 0.0045 for every rollout. Alternative constants (the mixture-mean-matched w=0.015 and the Δ=1-matched w=0.045) were run in the BF16 heterogeneous sweep (Section 3.2), where the $\Delta_{\max}$-anchored w=0.0045 performed best; only that point was carried into the FP4 heterogeneous experiments.

3. Results

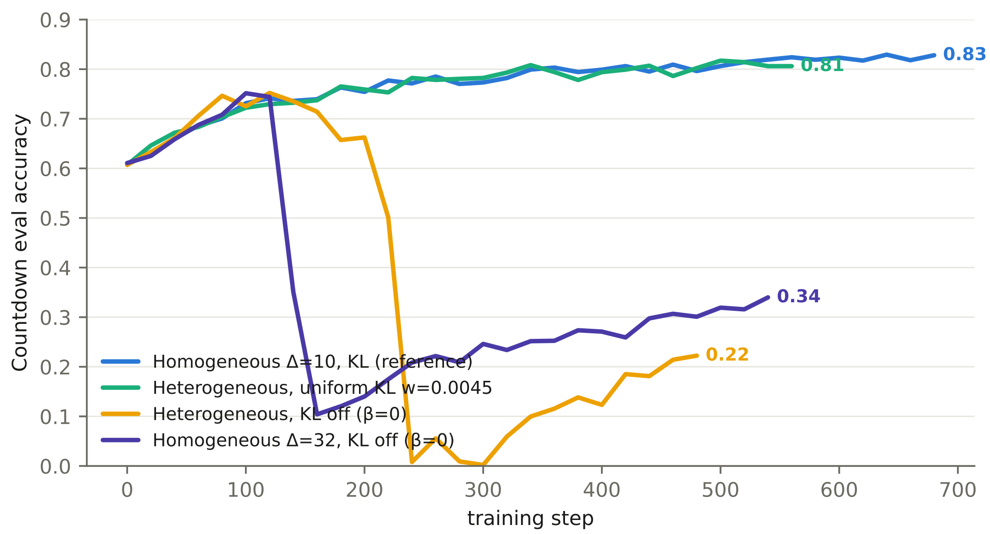

FIGURE 1. Collapse without the KL penalty; stability with it. Yellow (heterogeneous) and violet (homogeneous Δ=32) are β=0: both collapse (steps ∼230 and ∼170) and never recover. Aqua (uniform dose w=0.0045) trains through the same window untouched; blue is the homogeneous Δ=10 reference.

3.1. Does staleness alone break unregularized async RL?.

Yes — and the free surrogate prevents it. The homogeneous Δ=32, β=0 control peaks at eval 0.752 (step 100), then drops sharply between steps 160–180 (train reward 0.68 → 0.19; eval 0.14 at step 200) and oscillates around 0.2–0.45 for 400 more steps without recovering. The heterogeneous β=0 control — average lag 4× smaller — shows the same failure delayed by ∼60 steps, preceded by a characteristic train-reward overshoot (0.94 at step 127), then eval 0.002 by step 300. Every arm with the EMA-KL surrogate on, at any tested strength, trains stably through and far beyond these windows (Fig. 1). These runs are BF16 end to end: staleness alone, without quantization, breaks unregularized async RL at these lags, and the zero-cost surrogate fixes it.

As a control on the new machinery itself, the homogeneous Δ=10 reference run through the full pipeline (offsets mechanism, vLLM anchor, raw-units clamp) peaks at 0.829 / final 0.828 — inside the historical no-centering band (0.827–0.831) — so the infrastructure changes do not themselves shift the baseline.

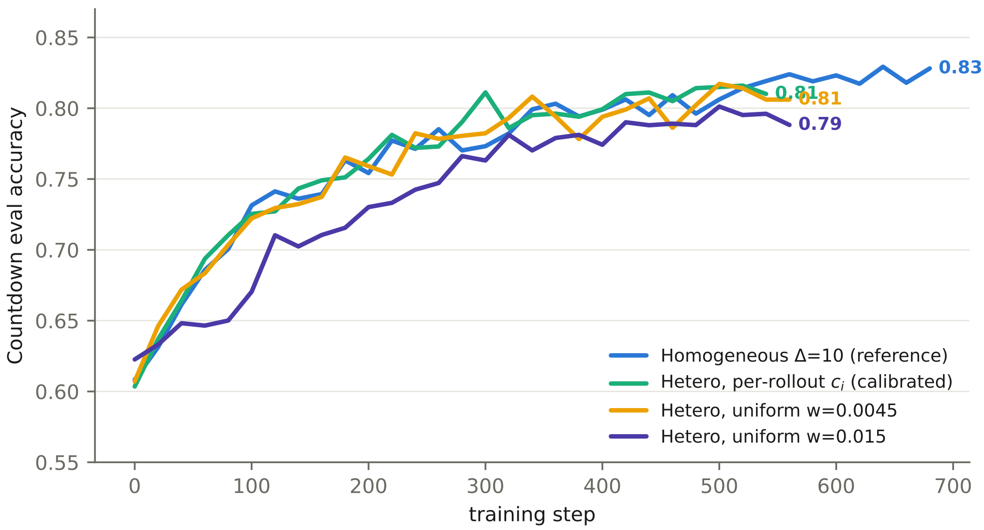

FIGURE 2. All stable arms. Calibrated per-rollout c_i (aqua) vs. the two uniform doses (yellow w=0.0045, violet w=0.015) under the same $\Delta_i \in \{1, 4, 10, 32\}$ mixture; blue is the homogeneous reference.

3.2. **Does per-rollout calibration beat a constant dose?** Not here — it ties the best constant, but the best constant is the one the theory picks. Peaks: calibrated 0.816 (step 520), uniform w=0.0045 0.817 (step 500), uniform w=0.015 0.801; homogeneous reference 0.829. The calibrated arm and the best uniform arm are statistically indistinguishable at one seed, and both sit ∼1 point under the reference. Two facts qualify this tie. First, the best uniform dose is exactly the one the theory prescribes ($w = \beta_{\text{ref}}c_{\text{ref}} = 0.0045$, transported from the known-good homogeneous run via the βc invariance), and tripling it costs 1.5 points: the calibration theory's practical payoff here is dose selection *without a sweep*, which is what a practitioner changing Δ (or mixing Δ_i) actually needs. Second, the experimental design gives per-rollout weighting little room to act. Because each K-group is policy-homogeneous (all K rollouts in a group share one Δ_i), switching from a constant to a per-rollout weight rescales each group's summed penalty by a common per-group factor, and the resulting cross-group differences in dose are diluted by batch-level normalization. Allocation of regularization across rollouts can therefore only matter when policy ages mix *within* a K-group — a configuration this batch does not contain, and the subject of ongoing work (Section 5).

3.3. **Is the first-order surrogate accurate everywhere?** No — its error is strongly Δ-dependent, and training does not care. Median relative error of means: 5.88 ($\Delta \le 2$), 2.13 ([3, 6]), 0.48 ([7, 20]), 0.19 (≥ 21) — the theory note's finite-difference variance warning, measured (Fig. 3). Despite a quarter of tokens carrying relative-error-∼6 estimates, the calibrated arm matched the reference; the summed, β-scaled penalty is robust to zero-mean estimator noise. This suggests a simple untested refinement: floor Δ_i at ∼4 in c_i.

3.4. **Does quantization compound with staleness?** Yes — and the same theory dose rescues it, fully at Δ=10 and partially at Δ=32. The β=0 ladder (Fig. 4): at Δ≈1, FP4 training is stable and strong over its 242-step run (train reward 0.80–0.85, eval 0.794 at step 220). At Δ=10 it survives its full horizon (∼420 steps), volatile in train reward but climbing in eval (0.802 at step 400). At Δ=32 it collapses by step ∼100 (train reward 0.38; eval already 0.17) and stays collapsed (0.303 at 400). The BF16 twin at the same lag collapsed at step ∼170 from a higher peak (0.752 vs. 0.637): quantization and staleness *compound* — the boundary moves ∼70 steps earlier and the pre-collapse peak drops — and at this horizon the β=0 boundary sits between Δ=10 and 32. The rescue arms apply the *same* theory dose βc = 0.0045 used throughout, with no quantization-specific retuning. At Δ=10 this is fully sufficient: healthy through the

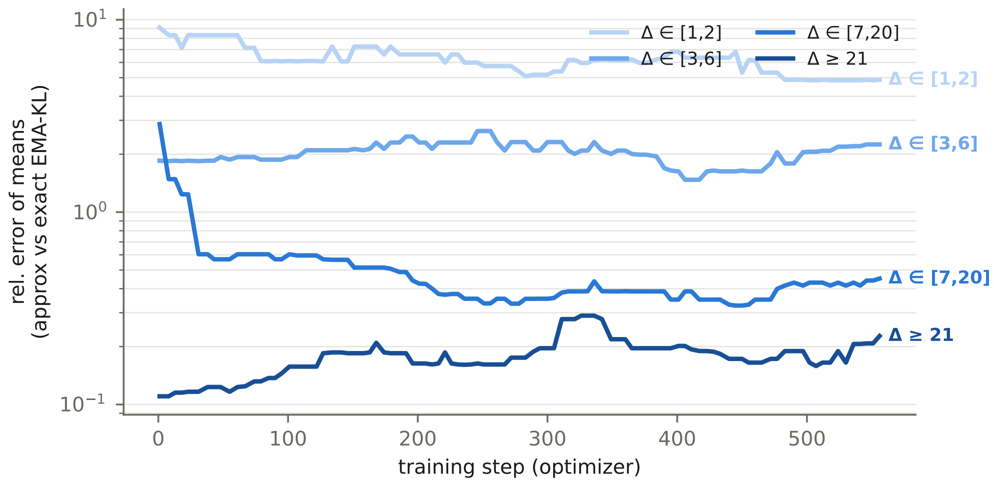

FIGURE 3. Relative error of the surrogate vs. exact EMA-KL by lag stratum (calibrated run; window-median smoothed; log scale).

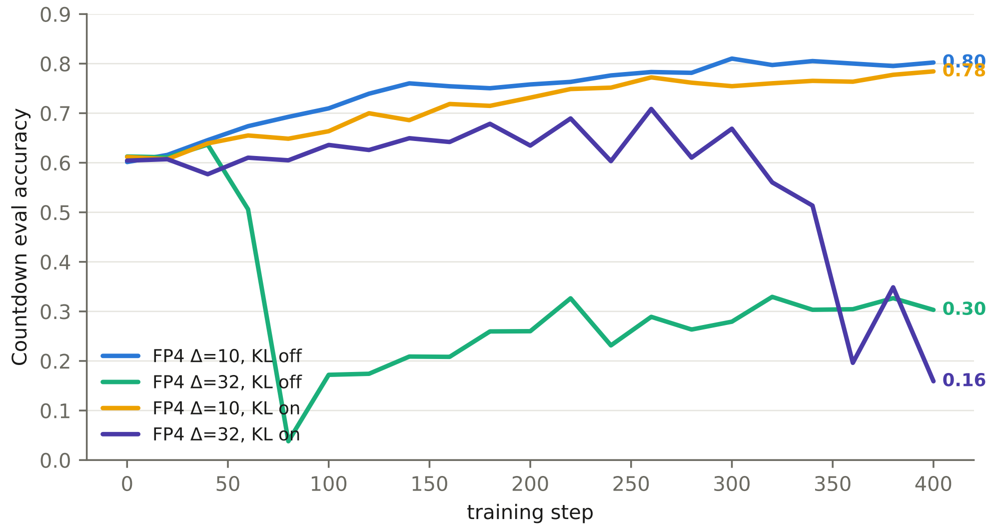

FIGURE 4. FP4 β=0 ladder and rescue arms. Blue/aqua: β=0 at Δ=10 (stable) and Δ=32 (collapses by step ∼100). Yellow/violet: the same theory dose $\beta c = 0.0045$ at Δ=10 (stable through the full horizon) and Δ=32 (holds ∼3× past the β=0 boundary, degrades late).

full horizon, eval 0.784 at step 400. At Δ=32 it carries training through and far beyond the twin's collapse window (0.636 at 100, 0.679 at 180, peak 0.709 at 260 — a ∼3× extension of the boundary), but the climb is slower and noisier at reduced altitude, and late in the run evals degrade (0.51 at 340 → 0.16 at 400): the unscaled dose delays rather than fully prevents the compounded failure. A quantization-aware dose scaling — treating the quantization gap as extra effective staleness — is the natural refinement, and is untested.

Negative result: the asymmetric arm fails, for an identifiable reason. The asymmetric arm (BF16 trainer, QDQ sampler, KL on at $\beta c = 0.0045$) degrades monotonically from the start: eval 0.56 at step 20, 0.04 at 100, 0.001 by step 180; the run was stopped at step 200. The mechanism is that the surrogate's persistent quantization gap enters the advantage-level (score-function) penalty un-centered, acting as a global negative reward that suppresses all sampled behavior rather than exerting corrective QAT pressure. A differentiable KL-in-loss path, or centering of the gap, would be needed for this setting.

FIGURE 5. The original-hypothesis cell: symmetric NVFP4 with heterogeneous staleness $\Delta_i \in \{1, 4, 10, 32\}$. Yellow ($\beta$=0) collapses at $\sim$step 205 and never recovers; both KL arms — per-rollout staleness-weighted c_i (blue) and uniform w=0.0045 (aqua) — train healthily through their full horizons and end within 0.007 eval of each other.

3.5. **Does staleness weighting matter under FP4 heterogeneity?** The rescue holds; weighted vs. constant is a tie at one seed. This is the cell that motivated the project: three arms, all symmetric NVFP4 QDQ with the heterogeneous offsets $\Delta_i \in \{1, 4, 10, 32\}$ at `async_level` 32 — a β=0 control, a uniform dose w=0.0045 (β=0.016, global Δ-source), and per-rollout calibrated c_i (β=0.005) (Fig. 5). The control collapses at $\sim$step 205 (train reward 0.75 at 140 $\to$ 0.51 at 200 $\to$ 0.26 at 212; eval 0.644 at 180 $\to$ 0.161 at 200) and never recovers through step 327 (eval 0.163 at 320). The boundary at $\sim$205 sits between the FP4-homogeneous Δ=32 collapse ($\sim$100) and the BF16-heterogeneous collapse ($\sim$230), consistent with an intermediate effective mismatch. Both KL arms trained healthily through their full horizons (final steps $\sim$361/381; last evals at step 360); notably, under the heterogeneous mixture the uniform dose did *not* reproduce the late ($\sim$340) degradation of its homogeneous-Δ=32 counterpart, presumably because the mixture's mean lag is milder. On eval, the two KL arms tie: peaks 0.771 (per-rollout, step 360) vs. 0.764 (uniform, step 340), finals 0.771 vs. 0.764 at step 360 — both gaps under 0.01 at a single seed, with the lead trading at matched steps (0.705 vs. 0.711 at 200; 0.756 vs. 0.741 at 300; 0.742 vs. 0.746 at 320). Train reward frequently favored the per-rollout arm (e.g. 0.807 vs. 0.735 at 200), but that lead traded on batch noise; we judge on evals. The confirmed part stands: staleness-weighted KL rescues FP4-heterogeneous training where β=0 collapses; the comparison against uniform dosing is a tie at this seed count, consistent with the age-homogeneous-group mechanism of Section 3.2.

3.6. **Does within-group age mixing separate weighting from a constant?** Yes — it ends the tie, though the two precisions disagree on which side wins. Section 3.2 argued that per-rollout weighting has no room to act while each K-group is policy-homogeneous, because switching from a constant to c_i then rescales each group's summed penalty by a common factor that batch-level normalization largely absorbs. The mechanism predicts a separation as soon as the ages mix *inside* a group, so we ran the 2×3 grid $\{$BF16, NVFP4$\} \times \{\beta$=0, uniform w=0.0045, per-rollout $c_i\}$ with within-group mixing and otherwise unchanged configuration. All six arms are still training; at the common floor of step 240 they stand at

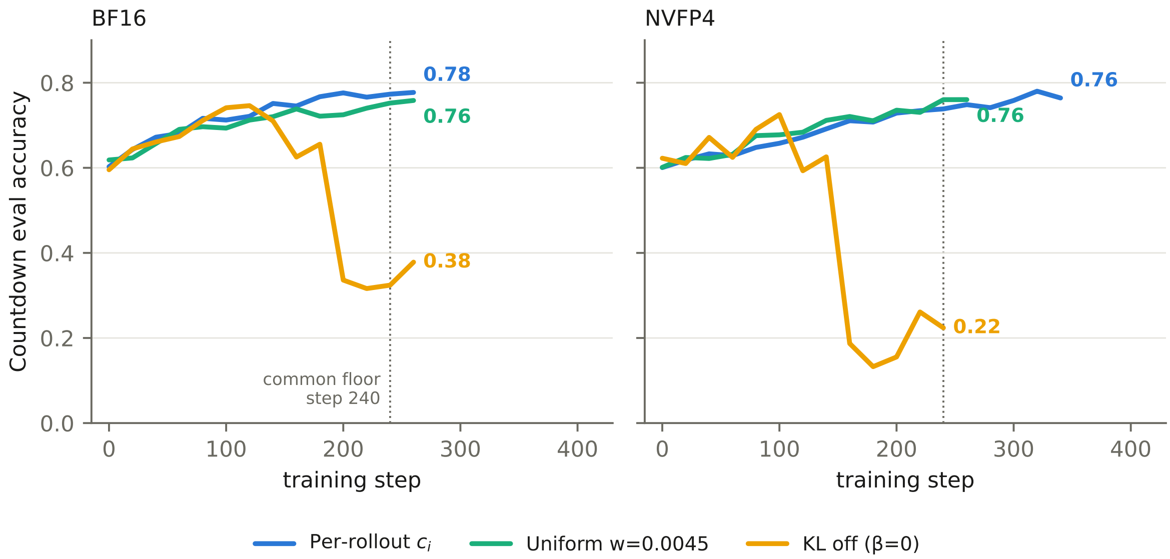

FIGURE 6. In progress. Within-group age mixing: every K-group spans $\Delta_i \in \{1, 4, 10, 32\}$. Same three arms, same colours, both precisions; dotted line marks the common step floor (240) at which the six arms are compared. β=0 (yellow) collapses in both panels, earlier under NVFP4; both KL arms train through those windows.

Within-group arm (eval at step 240)	BF16	NVFP4
Per-rollout c_i	0.773	0.738
Uniform w=0.0045	0.752	0.760
KL off (β=0)	0.324	0.224

Two things are already legible (Fig. 6). First, the rescue reproduces and the unregularized failure gets harder. Both β=0 controls collapse and neither recovers, and both collapse earlier than the identical lag multiset mixed *across* groups: ∼step 190 vs. ∼230 in BF16 (train reward 0.766 → 0.471 over steps 180–200, monitored $|\overline{\mathrm{KL}}|$ 14.6 → 297) and ∼150 vs. ∼205 under NVFP4 (train reward 0.686 → 0.377 over steps 140–160, $|\overline{\mathrm{KL}}|$ 30.8 → 699, with instability visible from step 120). This matters because within-group mixing is not an exotic configuration: it is what a throughput-maximizing scheduler produces by default once it stops pinning ranks to checkpoints. Holding the average lag fixed and only redistributing it inside the group costs ∼40–55 steps of stability, and the NVFP4 penalty on top of that (∼40 steps) is the same compounding measured in Section 3.4. Both KL arms train through these windows in both precisions, at the same dose used everywhere else in this report.

Second, the calibrated-versus-uniform comparison leaves the tie band — with opposite signs. Over the seven evals from step 140 to 260, the range both arms share in both precisions, per-rollout c_i leads the uniform dose by +0.029 eval on average in BF16 and is ahead at 7/7 of those evals; under NVFP4 the uniform dose leads by 0.010 and is ahead at 6/7. The same statistic on the age-homogeneous runs of Sections 3.2 and 3.5 is +0.004 (4/7) and +0.003 (3/7) — coin flips at a tenth the magnitude. The BF16 panel is the predicted result at roughly the predicted size; the NVFP4 panel is a smaller effect of the wrong sign, and we decline to read a ranking out of one seed per cell at a third of the intended horizon. What the grid does support is the weaker structural claim the mechanism actually makes, and which the earlier sections could not test: *allocation across rollouts is inert while groups are age-homogeneous and is not inert once they are not.* The total delivered penalty is not what changed — median $|\overline{\mathrm{KL}}|$ is 5.2 (per-rollout) vs. 1.6 (uniform) in these runs, against 5.1 vs. 1.5–1.9 in the between-group runs — so the two designs differ in where the weight is placed, not in how much is delivered. Deciding the sign requires the full horizons and ≥ 3 paired seeds.

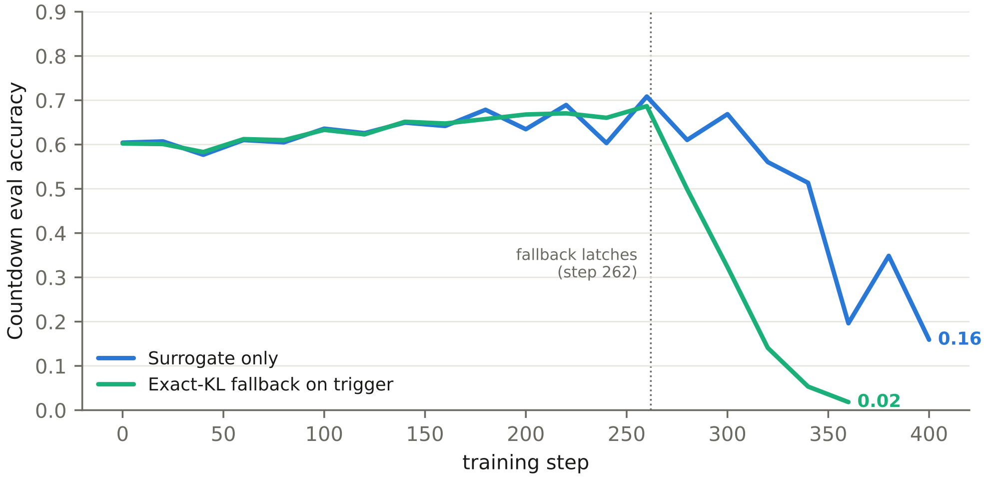

FIGURE 7. In progress. FP4 Δ=32, $\beta c = 0.0045$, with and without the error-triggered exact-KL fallback. The two runs are indistinguishable until the fallback latches at step 262 (dotted), after which the switched arm falls away 60–80 steps ahead of its twin.

3.7. **Can an error-triggered exact-KL switch rescue Δ=32?** No — the alarm is sound and the intervention is harmful. The FP4 Δ=32 rescue arm of Section 3.4 degrades late, and its surrogate diagnostics degrade first: in that run the relative error of means sits at 1.03–1.07 from step 60 through 200, then reaches 1.6 at 240, 4.3 at 260 and 23.6 at 300 — 80–100 steps ahead of the eval turn near step 340 — while the bias stays negative and grows ($-0.05 \to -0.16$): the surrogate under-doses exactly when the brake is needed. The corresponding intervention is free, because the exact EMA-KL term is already computed every step for these diagnostics: a run can watch $|\overline{\mathrm{KL}}|$ against its own steps-100–200 baseline and, on a sustained excursion (2× for 20 consecutive rollout steps), latch the loss to the exact branch for the remainder of training. We ran the same configuration with that fallback armed.

The detector behaved exactly as designed. It fired at rollout step 262 ($|\overline{\mathrm{KL}}| = 13.5$ against a baseline of 4.54), ahead of any eval turn — eval was at its run maximum, 0.687 at step 260. Up to that point the fallback run and its surrogate-only twin are indistinguishable (mean paired eval difference +0.003 over steps 0–260, never exceeding 0.06 in magnitude), which is what makes the divergence afterwards attributable to the switch rather than to seed noise: eval 0.499 (280), 0.323 (300), 0.141 (320), 0.053 (340), 0.018 (360), against the twin's 0.610, 0.669, 0.560, 0.514, 0.196 — a paired deficit that widens at every step to 0.46 by step 340, after which the twin begins failing too (Fig. 7). The twin's own turn comes near step 360; the fallback moves that failure ∼60–80 steps earlier and drives it an order of magnitude deeper.

The diagnostics suggest the switch is also, inadvertently, a dose change. After latching, the delivered penalty runs at 3–7× its own pre-trigger baseline ($|\overline{\mathrm{KL}}|$ 14.9–33.6 vs. 4.54) and the gradient norm rises from ∼0.06 to 0.17–0.47. A mechanism is available: the two estimators are truncated in different units — the surrogate clamps the raw log-ratio at ±10 before multiplying by $c = 0.28125$, whereas the exact term clamps the KL itself — so at fixed β they are not interchangeable once drift is large, and swapping them mid-run raises the effective strength at the least forgiving moment. We flag this as a hypothesis, not a result: with one seed we cannot separate the post-latch growth in penalty magnitude from the collapse it accompanies. The usable conclusion is the conservative one. A zero-cost accuracy monitor is worth keeping as an alarm — it selected the right run and the right moment, firing while eval was still at its maximum — but estimator substitution is not a free correction. Any future version should be dose-matched at the latch, rescaling β so the delivered penalty is continuous across the switch, and should be evaluated against a same-seed control that latches to the surrogate itself.

4. Related work

EMA anchors in RL for LLMs. EMA-PG [2] regularizes policy-gradient training of LLMs against an *explicit* EMA anchor, maintained as a separate model with its own reference forward pass, and treats asynchrony as an orthogonal concern. Our setting obtains an EMA anchor *implicitly*, at zero cost, precisely because of asynchrony: the stale sampler logprobs are already cached, and the theory note [1] shows they support a first-order surrogate for the EMA-KL term. In alignment, TR-DPO [3] and WARP [4] use sequence-level EMA (or averaged) reference policies, updating the anchor toward the trainee during training; these share the moving-anchor intuition but pay for explicit reference evaluations.

KL to the inference policy in async RL. Kimi-K2-style training [5] penalizes divergence from the inference (sampling) policy in an asynchronous pipeline — exactly the object our theory identifies as implicit EMA regularization when the coefficient is held constant. The truncated-importance-sampling line [6] addresses the same trainer/sampler mismatch through the importance-weight path rather than a penalty; the two mechanisms are complementary, and all our arms run with importance sampling on.

KL-free recipes and FP4 training. DAPO [8] removes the KL term entirely, and the humans& NVFP4 recipe [7] trains KL-free (the miles trainer defaults `kl_coef=0`), managing trainer/sampler mismatch through systems-level bit-exact kernel contracts. Our results suggest the free surrogate composes with such recipes: it costs nothing, and it converts high-staleness collapse into stable training. On the numerics side, NVFP4 pretraining [9] and Four-Over-Six [10] develop the quantization formats and scaling rules our QDQ emulation follows.

5. Future work

Five directions follow directly from the findings. (1) *Resolving the within-group sign*: Section 3.6 shows that mixing ages inside a K-group does move calibrated dosing off the tie with a constant, but BF16 and NVFP4 disagree on the direction at a third of the horizon. This is the one place where a ranking is plausibly obtainable, and it needs the full 720 steps and ≥ 3 paired seeds per cell rather than more configurations. (2) *Dose-matched estimator switching*: the fallback of Section 3.7 shows that swapping the surrogate for the exact term at fixed β changes the delivered strength; the corrected version rescales β at the latch so the penalty is continuous, and is compared against a same-seed control that latches to the surrogate. (3) *Quantization-aware dose scaling*: the late degradation of the FP4 Δ=32 rescue suggests treating the quantization gap as extra effective staleness when setting the dose — an alternative to (2) that raises the dose in advance rather than in reaction. (4) *The asymmetric setting*: a gap-centered variant or a differentiable KL-in-loss path, so the quantization gap exerts corrective rather than globally negative pressure. (5) *A Δ_i floor* ($\sim$4) in c_i, suggested by the stratified error of Section 3.3; the within-group runs are the natural place to test it, since they are the first configuration in which small-Δ rollouts and large-Δ rollouts share a group and their weights therefore compete.

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Appendix A. Configurations

Common: `model=Qwen/Qwen3-4B-Instruct-2507`, `LR=1e-6`, `sampled_k=train_k=16`, `steps=720`, `type=tb`, `conf=COUNTDOWN.toml`, `seq_length=4096`, `eval_steps=20`, `reference_mode=ema`, `ema_alpha=0.9`, `kl_centering=false`, `vllm_logprobs=true`, batch 512, wandb `fp4_KL`, branch `fp4_KL`.

Arm	β	async	offsets	Δ-source
`homog10_approxKL_b005_vllmLP`	0.005	10	10,10,10,10	per_rollout
`hetero_approxKL_b005_perRollout`	0.005	32	1,4,10,32	per_rollout
`hetero_approxKL_global_w0045`	0.016	32	1,4,10,32	global
`hetero_approxKL_global_w015`	0.0533	32	1,4,10,32	global
`hetero_b0_async32`	0	32	1,4,10,32	—
`bf16_b0_async32`	0	32	— (legacy rule)	—
`fp4sym_smoke` (60 steps)	0.005	10	—	global
`fp4sym_b0_async1`	0	1	—	—
`fp4sym_b0_async10`	0	10	—	—
`fp4sym_b0_async32`	0	32	—	—
`fp4sym_approxKL_b005_async10`	0.005	10	—	global
`fp4sym_approxKL_b0016_async32`	0.016	32	—	global
`fp4asym_approxKL_b005_async10`†	0.005	10	—	global
`fp4hetero_b0_async32`	0	32	1,4,10,32	—
`fp4hetero_approxKL_global_w0045`	0.016	32	1,4,10,32	global
`fp4hetero_approxKL_b005_perRollout`	0.005	32	1,4,10,32	per_rollout
`fp4sym_b0016_async32_klFallback`§	0.016	32	—	global
`heteroWG_b0`	0	32	1,4,10,32*	—
`heteroWG_global_w0045`	0.016	32	1,4,10,32*	global
`heteroWG_perRollout`	0.005	32	1,4,10,32*	per_rollout
`fp4heteroWG_b0`	0	32	1,4,10,32*	—
`fp4heteroWG_global_w0045`	0.016	32	1,4,10,32*	global
`fp4heteroWG_perRollout`	0.005	32	1,4,10,32*	per_rollout

KL penalty: $-\beta \sum_t c\,\mathrm{clamp}(\log T_n - \log T_{n-\Delta}, \pm 10)$ added to advantages (no centering); c per-token (c_i) or global per the table. FP4 arms add `rollout_quant=nvfp4` + `fake_quant=nvfp4` (252 patched linears; embeddings, `lm_head`, norms BF16). †BF16 trainer, QDQ sampler (asymmetric; `fake_quant` off). §Adds `kl_fallback=exact_on_trigger` ($2\times$ over the steps-100–200 $|\overline{\mathrm{KL}}|$ baseline, sustained 20 rollout steps); otherwise identical to `fp4sym_approxKL_b0016_async32`. *Lags mixed *within* each K-group (`staleness_within_group=true`) rather than across data-parallel ranks. The last seven rows are runs in progress.

Appendix B. Eval table (selected steps)

Step	100	200	300	400	500	final
Homog. Δ=10 reference	0.731	0.754	0.773	0.799	0.806	0.828 (680)
Hetero, per-rollout c_i	0.725	0.764	0.811	0.799	0.815	0.810 (540)
Hetero, uniform w=0.0045	0.722	0.759	0.782	0.794	0.817	0.806 (560)
Hetero, uniform w=0.015	0.670	0.730	0.763	0.774	0.801	0.788 (560)
Hetero, β=0	0.725	0.662	0.002	0.123	—	0.222 (480)
Homog. Δ=32, β=0	0.752	0.140	0.246	0.271	0.319	0.340 (540)
FP4 $\Delta\approx$1, β=0‡	0.726	0.786	—	—	—	0.779 (240)
FP4 Δ=10, β=0	0.710	0.758	0.810	0.802	—	0.802 (400)
FP4 Δ=32, β=0	0.172	0.260	0.279	0.303	—	0.303 (400)
FP4 Δ=10, KL on	0.664	0.731	0.755	0.784	—	0.784 (400)
FP4 Δ=32, KL on	0.636	0.635	0.669	0.159	—	0.159 (400)
FP4 asym. (BF16 trainer)	0.036	—	—	—	—	0.001 (180)
FP4-hetero, β=0‡	0.716	0.161	0.254	—	—	0.163 (320)
FP4-hetero, uniform w=0.0045	0.659	0.711	0.741	—	—	0.764 (360)
FP4-hetero, per-rollout c_i	0.649	0.705	0.756	—	—	0.771 (360)
In progress (Sections 3.6, 3.7); last column is the last completed eval.						
Within-group BF16, per-rollout c_i	0.712	0.776	—	—	—	0.777 (260)
Within-group BF16, uniform w=0.0045	0.693	0.725	—	—	—	0.758 (260)
Within-group BF16, β=0	0.741	0.336	—	—	—	0.378 (260)
Within-group FP4, per-rollout c_i	0.658	0.729	0.758	—	—	0.764 (340)
Within-group FP4, uniform w=0.0045	0.678	0.736	—	—	—	0.760 (260)
Within-group FP4, β=0	0.725	0.155	—	—	—	0.224 (240)
FP4 Δ=32, KL on + fallback	0.633	0.668	0.323	—	—	0.018 (360)

‡Run ended early by an infrastructure fault. Rows below the rule are truncated at the last completed eval of a run still training; cross-arm comparisons in the text are made at a common step floor, not at these final values.